Compiled Solutions – Senior Division

1. 01-02 C1 Computer Number Systems

$X_8 = BAD_{16} - 465_8 + 447_8 = 5655_8 - 465_8 + 447_8 = 5637_8$

5637_8 or 5637

2. 01-02 C1 Computer Number Systems

The indicated sum is $B3CAE_{16}$. This converts to: 10110011110010101110_2. The number of 1's is 12.

12

3. 01-02 C2 Computer Number Systems

$F123_{16} = 1111000100100011_2 = 170443_8$ so $170443_8 - 123_8 = 170320_8$
$123_{10} = 1 * 8^2 + 7 * 8 + 3 = 173_8$ so $170320_8 - 173_8 = 170125_8 =$
$1\ 111\ 000\ 001\ 010\ 101_2$
Converting all values to hexadecimal gives $F123_{16} - 7B_{16} - 53_{16} = F055_{16}$
Converting to binary gives $1111\ 0000\ 0101\ 0101_2$

1111000001010101_2
or
1111000001010101

4. 02-03 C1 Computer Number Systems

$1_{10} = 1 = 3^0$
$11_2 = 3 = 3^1$ $11_8 = 9 = 3^2$ $33_8 = 27 = 3^3$
$1010001_2 = 81 = 3^4$ $363_8 = 243 = 3^5$ $3^6 = 729 = 2D9_{16}$

2D9 or $2D9_{16}$

5. 02-03 C1 Computer Number Systems

Start checking for the smallest 3-digit hex number by using 1 as the first digit. 92B + 1 * * = A * *. The last digit of the sum must now end in A. 92B + 1 * F = A * A. The middle digit must be a 0. 92B + 10F = A3A.

10F or $10F_{16}$

6. 02-03 C2 Computer Number Systems

$(FF_{16})^2 = (100_{16} - 1_{16})^2 = 100^2 - 2*100 + 1^2 = 10000_{16} - 200_{16} + 1 = FE01_{16}$

FE01 or $FE01_{16}$

7. 03-04 C1 Computer Number Systems

A. $431_8 - 11_{16} = 100001000$
B. $1493_{10} + 72_8 = 11000001111$
C. $4B7_{16} + 32_8 = 10011010001$
D. $101_2 + 101_8 = 1000110$
E. $B21_{16} - 72_{10} = 101011011001$
B and E have more 1's than 0's.

B and E

8. 03-04 C1 Computer Number Systems

$CA_{16} / AC_{16} = (12*16+10) / (10*16+12) = 202 / 172 = 1\ r1E_{16} = 11\ 110_2 = 36_8$

36 or 36_8

9. 03-04 C2 Computer Number Systems

$4_{10} * 1011_2 = 1011 + 1011 + 1011 + 1011 = 101100_2 = 44_{10}$ -144
$3_{10} * 126_8 = 126 + 126 + 126 = 402_8 = 258_{10}$
$2_{10} * DF_{16} = DF + DF = 1BE_{16} = 446_{10}$
$44 + 258 - 446 = -144_{10}$

10. 04-05 C1 Computer Number Systems

$2 * X_{16} + 11011010_2 = 4320_8$ 3FB or $3FB_{16}$
$4320_8 = 100011010000_2$
$2 * X_{16} = 100011010000_2 - 11011010_2 = 11111110110_2$
$X_{16} = 11\ 1111\ 1011_2 = 3FB_{16}$

11. 04-05 C1 Computer Number Systems

D has the most 1's in its binary representation? D
A) $2345_8 = 10011100101_2$ 6 1's
B) $FAB_{16} = 111110101011_2$ 9 1's
C) $511_{10} = 111111111_2$ 9 1's
D) 1011011101111_2 10 1's
E) $3794_{16} = 11011110010100_2$ 8 1's

12. 04-05 C2 Computer Number Systems

$31_8 * X_2 + 12_{16} = 28_{16} * X_2 - 1001000_2$ converting to base 10 gives: 110 or 110_2
$25 * X_2 + 18 = 40 * X_2 - 72$
$90 = 15 * X_2$
$6 = X_2 = 110_2$

13. 05-06 C1 Computer Number Systems

Converting to base 10, $(X_{16})^2 - 1010101_2 * X_{16} + 1356_8 = 0$ A, 4B or A_{16}, $4B_{16}$
becomes $X^2 - 85X + 750 = 0$
$(X - 10)(X - 75) = 0$
$X = 10$ OR $X = 75$
$X_{16} = A$ or $X_{16} = 4B$

14. 05-06 C1 Computer Number Systems

$504_8\ X_2 - 2E6_{16} = 80_{10}\ X_2 + 11101010_2$ 100 or 100_2
$324_{10}\ X_2 - 742_{10} = 80_{10}\ X_2 + 234_{10}$
$244_{10}\ X_2 = 976_{10}$
$X_2 = 4_{10} = 100_2$

15. 05-06 C2 Computer Number Systems

$10010_2 = 18$, $20_{16} = 32$, $31_8 = 25$, $11001_2 = 25$, $25_{10} = 25$ and $19_{16} = 25$ — 11001 or 11001_2
The mean = 150/6 = 25 = 11001_2

16. 06-07 C1 Computer Number Systems

$F_{16} = 10_{16} - 1_{16}$, $FF_{16} = 100_{16} - 1_{16}$ ……. — 111110A / $111110A_{16}$
Adding the terms gives $1111110_{16} - 6_{16} = 111110A_{16}$

17. 06-07 C1 Computer Number Systems

$24_8 * X_2 - AC_{16} = 101_2 * X_2 - 235_8$ — 1 or 1_2
$20_{10} * X_2 - 172_{10} = 5_{10} * X_2 - 157_{10}$
$15_{10} * X_2 = 15_{10}$
$X_2 = 1_{10} = 1_2$

18. 06-07 C2 Computer Number Systems

$367_8 + 10111_2 = X_{16} - A12_{16}$ — B20 or $B20_{16}$
$367_8 + 27_8 = X_{16} - 101000010010_2$
$416_8 = X_{16} - 101000010010_2$
$X_{16} = 100001110_2 + 101000010010_2 = 101100100000_2 = B20_{16}$

19. 07-08 C1 Computer Number Systems

$101_2 = 5$ $32_8 = 26$ $24_{16} = 36$ $111_2 = 7$ — 235 or 235_{16}
The expression in base 10 is: 5 * 26 + 15 (36 – 7) = 565 = 235_{16}

20. 07-08 C1 Computer Number Systems

$1X6_{16} = 256 + 16X + 6 = 262 + 16X$ — 5
$X26_8 = 64X + 16 + 6 = 64X + 22$
Therefore, $64X + 22 = 16X + 262$
$48X = 240$
$X = 5$

21. 07-08 C2 Computer Number Systems

The smallest 3-bit palindrome is 101 and the largest is 777. — 2
101 is 65 in base 10 and 777 is 511
$81_{10} = 121_8$ and $121_{10} = 171_8$ so there are 2 of them.

22. 08-09 C1 Computer Number Systems

$L = 31_8 = 25$ and $W = 10110_2 = 22$ and $H = 3F_{16} = 63$ — 875A or $875A_{16}$
Therefore, $V = 25 * 22 * 63 = 34650 = 875A_{16}$

23. 08-09 C1 Computer Number Systems

$10_2 * X + 2AE_{16} = 4710_8$ — 38D or $38D_{16}$
Converting to decimal yields
$2 * X + 686 = 2504$
$2 * X = 1818$
$X = 909_{10} = 38D_{16}$

24. 08-09 C2 Computer Number Systems

First convert all to base 10: — A and B
A. 292_{10} $= 292$
B. $113_{16} + 11_{16}$ $= 124_{16} = 292$
C. 100100110_2 $= 294$
D. $513_8 - 46_8 = 445_8$ $= 293$

25. 09-10 C1 Computer Number Systems

$22_8 = 18$
$1011101_2 = 93$
$20_{16} = 32$
$67_8 = 55$

So $22_8 * X_{16} – 1011101_2 = 20_{16} + 67_8$
becomes $18* X – 93 = 32 + 55$
$18 * X - 93 = 87$
$18 * X = 180$
$X_{10} = 10 \Rightarrow X_{16} = A$

A or A_{16}

26. 09-10 C1 Computer Number Systems

Since the numbers are so large, it is easier to compute in hex**.** — 15ACE or $15ACE_{16}$
$FEED_{16} + DEAF_{16} = 1DD9C_{16}$ $1DD9C_{16} - BEAD_{16} = 11EEF_{16}$
$11EEF_{16} + FACE_{16} = 219BD_{16}$ $219BD_{16} - BEEF_{16} = 15ACE_{16}$

27. 09-10 C2 Computer Number Systems

$BE + EB = 1A9_{16}$ $1A9 + 9A1 = B4A_{16}$ $B4A + A4B = 1595_{16}$ — 6EE6 or $6EE6_{16}$
$1595 + 5951 = 6EE6_{16}$

28. 10-11 C1 Computer Number Systems

$ACED_{16} = 126355_8$ $BEAD_{16} = 137255_8$ — DEED or $DEED_{16}$
$CAFÉ_{16} = 145376_8$ $DEED_{16} = 157355_8$
$FACE_{16} = 175316_8$
So DEED has the most which is 3.

29. 10-11 C1 Computer Number Systems

$21_8 = 11_{16}$ and $424_8 = 114_{16}$ 17
So there are 17 hexadecimal palindromes which are:

11	22	33	44	55	66	77	88	99
AA	BB	CC	DD	EE	FF	101	111	

30. 10-11 C2 Computer Number Systems

1010_2 N = 10 N 74_8 S = 60 S $5A_{16}$ N = 90 N 32_{16} N
1100100_2 S = 100 S 24_8 S = 20 S 32_{16} N = 50 N
1010000_2 N = 80 N If N is positive and S is negative,
then $10 – 60 + 90 – 100 – 20 + 50 + 80 = 50 \text{ N} = 32_{16}$ N

31. 11-12 C1 Computer Number Systems

There are 19 binary representations to consider. 63
Twelve start with 0010 and 7 start with 0011. Now analyze the 5th and 6th bits: 7 are 01, 4 are 10, 4 are 00, and 4 are 11. Looking at the last 2 yields: 5 are 00, 5 are 01, 5 are 10, 4 are 11. This makes a total of 63.

32. 11-12 C1 Computer Number Systems

$143_X = 11*X + 11$ expands to: x = 8, -1
$X^2 + 4 * X + 3 = 11* X + 11$
$X^2 – 7 * X – 8 = 0$
$(X + 1)(X – 8) = 0 \Rightarrow X = 8$ or X = -1

33. 12-13 C1 Computer Number Systems

$170_8 = 120$ $144_{10} = 144$ $1110000_2 = 112$ 203 or 203_8
$97_{16} = 151$ $200_8 = 128$
Sum = 655 Mean = 131 = 203_8

34. 12-13 C1 Computer Number Systems

$1011011_2 = 91$ $7321_8 = 3793$ 73B or $73B_{16}$
So $1011011_2 + 2 * X = 7321_8$ becomes
$91 + 2 * X = 3793$
$2 * X = 3702$
$X = 1851 = 73B_{16}$

35. 13-14 C1 Computer Number Systems

$2_{10} * X_{16} - 464_8 = 11101110_2 - X_{16}$ B6 or $B6_{16}$
$2_{10} * X_{16} - 308_{10} = 238_{10} - X_{16}$
$3_{10} * X_{16} = 546_{10}$
$X_{16} = 182_{10} = B6_{16}$

36. 13-14 C1 Computer Number Systems

328 = 101001000 332
329 = 101001001
330 = 101001010
331 = 101001011
332 = 101001100

37. 14-15 C1 Computer Number Systems

a) $11001010_2 + 13_8 = 202 + 11 = 213$ e
b) $257_8 + 24_{16} = 175 + 36 = 211$
c) $B4_{16} + 11110_2 = 180 + 30 = 210$
d) $128 + 23_8 + 101010_2 + 17_{16} = 128 + 19 + 42 + 23 = 212$
e) $136 + 1001110_2 = 136 + 78 = 214$

38. 14-15 C1 Computer Number Systems

$2_{10} * 416_8 + A23_{16} - 10110_2 + 31_8 / 101_2$ 6056 or 6056_8
$= 2 * 270 + 2595 - 22 + 25 / 5$
$= 540 + 2595 - 22 + 5$
$= 3118 = 6056_8$

39. 15-16 C1 Computer Number Systems

$15D_{16} < X_{10} + 1010_2 < 1271_8$ 340
Converts to: $349 < X + 10 < 697$
$339 < X < 687$
Therefore, the smallest integral value of X is 340.

40. 15-16 C1 Computer Number Systems

1_2 100_2 20_8 100_8 100_{16} 400 or 40016
1 4 16 64 256
Rule: The powers of four so $4^5 = 1024 = 400_{16}$